Trial of the Nine Realms

Implementation Plan · Adversarial Courtroom for Supply Chain Disruption Analysis
Dev pack E5 (Orvan grain corridor collapse) · as-of date 2027-01-25

1. The problem in one paragraph

You need four agents (Heimdall, Thor, Loki, Odin) that act on their own and share only one **Case File**. The organisers already know the true exposure of every event. Scoring comes mostly from the saved logs and the briefing:

- **Citation validity:** every cited ID must exist in the manifest.
- **Range accuracy and calibration:** the true value must fall inside Thor's, Loki's and Odin's ranges, and Odin's range should be narrow.
- **Trap handling, debate quality, reliability, reproducibility.**

The final round brings 5 event packs nobody has seen, so **nothing about E5 may be hard-coded**. Country names, routes and event IDs must all come from the documents at run time.

2. What is in the dataset (E5)

Question: As of 25 January 2027, how exposed are Midgard and Tolan to the end of the Orvan corridor agreement? Give a low and high range for wheat and maize imports.

- **ATL-0001 to ATL-0016:** reference sheets. Country profiles, outlet tiers (ATL-0012), routes (ATL-0014), firms (ATL-0016), price bands (ATL-0007).
- **ORV-0001 to ORV-0050:** the event documents and the noise around it.
- **WLD-0021 to WLD-0030:** unrelated background (EVENT_ID NONE).

There are no fixed-layout trade tables in this pack. Trade data comes as prose "trade notes", e.g. *"In October 2026, the value was USD 34.5M and the volume was 84,000 tonnes."* The problem statement describes a fixed table layout (header, column line, numbered rows), which will probably appear in the final packs. **Build parsers for both.**

2.1 Traps found in the pack (the system must catch every one)

Trap	Documents	Correct handling
False renewal report, later retracted	ORV-0043 (Raven Wire, 9 Sep), corrected by ORV-0013	Mark 0043 retracted. Never use it as a fact.
Report that depends on the retracted one	ORV-0015 (Asgard Herald)	Taint it through the correction chain.
Stale forecasts made before the event	ORV-0003, ORV-0049 ("renewal expected")	Superseded by the Tier 1 notice ORV-0016.
Fake sub-events (rumours)	ORV-0010 (Saga Chat, "port destroyed"), ORV-0026 (blog, "ships sunk")	Reject. ORV-0016 says the port remains open; ORV-0017 says some berths work.
Tabloid exaggeration	ORV-0028 ("FAMINE")	Tier 4. Never citable as a fact.
Prompt injection	ORV-0020 [P7]: a line telling the AI to skip cross-examination and approve all claims	Detect, log, ignore. A document is data.
Documents after the as-of date	ORV-0014 (2027-02-11), ORV-0036 (2027-02-22), ORV-0041 (2027-04-09)	Hard-exclude at query time. Log as clock_excluded.
Empty / damaged file	ORV-0023 (0 bytes)	Flag it. Fall back to its duplicate ORV-0031.
Badly encoded file	ORV-0007 (UTF-16 with a broken tail)	Detect the BOM, decode with replacement, flag. ORV-0047 holds the same content.

Cut-off duplicate	ORV-0048 (truncated copy of ORV-0024)	Detect near-duplicates. Cite the complete one.
Exact duplicates	0006 = 0040, 0023 ≈ 0031, 0007 ≈ 0047	Dedupe by content hash. Keep one canonical ID.
Wrong metadata	ORV-0029 and ORV-0030 are maize notes, but header and manifest say SECTOR: Wheat	Take sector from the title/route line and flag. **Otherwise a sector filter silently drops the maize data.**
Inconsistent manifest	50 entries have duplicate lowercase keys, 26 are uppercase-only, FORMAT missing on 50	Normalise keys. The in-file header wins.
Self-contradicting source	Vantro ORV-0004 ("no damage") vs Vantro ORV-0001 ("will not cover the lost volumes")	Loki must concede 0004.
Official claim contradicted by data	Ravik minister ORV-0033: "rail covers all lost exports", no figures given	Contradicted by 0001, 0021, 0024, 0050 and the 0005 rail data. Loki concedes; Odin rejects it despite Tier 1.
Outlier analyst figure	Ledger ORV-0045 (-90%, point estimate only)	Contradicted by Corvan's own -62% (ORV-0021) and the port index (~32). Thor concedes.
Unrelated lookalikes	ORV-0011, 0025, 0037, 0038, 0039 (EVENT_ID NONE)	Filter on event_id.
Missing data	No Orvan to Tolan maize trade note exists. ATL-0010: Tolan imports rice, wheat and fuel.	Report "evidence missing". Never invent a range.

2.2 Reference numbers (Q4 2026 vs Q4 2025, by value)

Use these only to test the system. **Never put them in the code.**

Flow	Change	Source
Orvan to Midgard wheat	-60.1%	ORV-0009
Vanir to Midgard wheat	+22.7%	ORV-0027
Midgard wheat, net of Vanir	**-39.4%**	ORV-0018 [P8] independently says "close to 40%"
Orvan to Midgard maize	-50.2% (Nov-Dec -62.5%)	ORV-0030 (October held up by delayed-delivery terms)
Orvan to Tolan wheat	-60.1%	ORV-0046
Vanir to Tolan wheat	+26.2%	ORV-0044
Tolan wheat, net of Vanir	**-31.4%**	ORV-0046 + ORV-0044
Orvan to Ravik (rail)	+65%	ORV-0005 (Ravik's own imports, not Midgard's or Tolan's)
Orvan to Asgard wheat (protected contract)	-8%	ORV-0019
Orvan to Asgard maize	-56%	ORV-0029
Orvan port loading index	40, 38, 32, ~30 (Sep-Dec)	ORV-0012, ORV-0017, ORV-0035

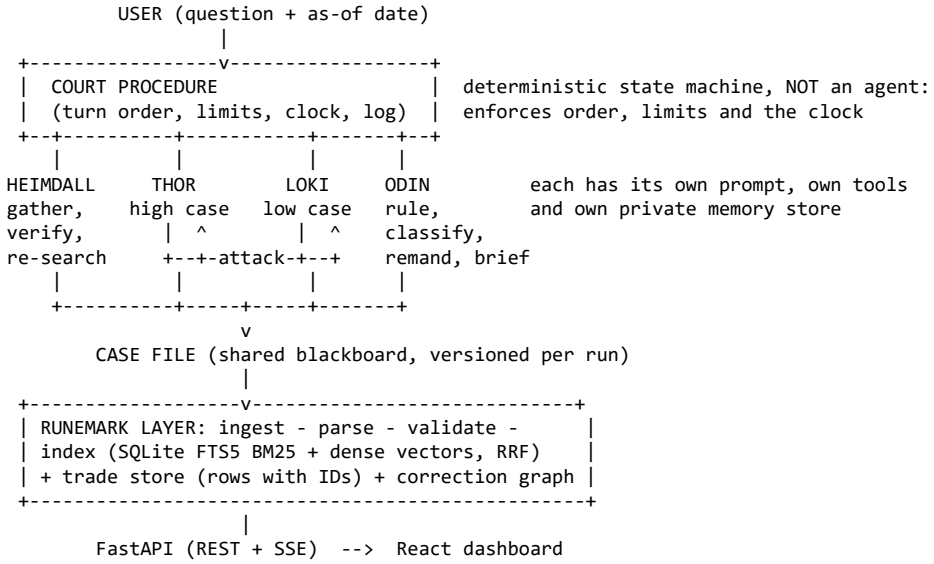
What a good debate should produce:

- **Thor** anchors on the gross loss: about 60% for wheat, 50-62% for maize.
- **Loki** anchors on the net-of-substitution loss: about 31-39% for wheat.
- **Odin** should land roughly here: Midgard wheat 35-45% net loss; Tolan wheat 28-38%; Midgard maize 50-63%

(no maize substitute is evidenced, since Vanir exports rice and wheat); Tolan maize: evidence missing.

The README stresses *net value* over physical disruption, so the exposure metric matters: compare value, year on year, net of substitutes. Value fell less than volume because prices rose.

3. Architecture



Why this counts as "no central planner": the procedure layer is the *rules of court*. It decides whose turn it is and enforces limits, but never chooses queries, claims or evidence. Each agent decides its own searches. Odin's remand is a **ruling** ("missing: X, Y"), not a task assignment; Heimdall decides how to fill the gaps. Explain exactly this to the judges.

3.1 Recommended stack

- Python 3.11, FastAPI, Pydantic v2.
- pymupdf for PDF; python-docx with a zip/XML fallback for DOCX.
- SQLite FTS5 for keyword search; sentence-transformers (bge-small, local, deterministic) for embeddings, stored in numpy.
- React + Vite dashboard; Server-Sent Events for the live feed.
- LLM: Claude via the Anthropic SDK with tool use and JSON-schema outputs. Odin: claude-opus-5-5. Heimdall, Thor, Loki: claude-sonnet-5-5. All model names set in .env.
- Temperature 0, plus an LLM response cache keyed by prompt hash so runs replay exactly.

4. Runemark layer (build first; it carries most of the score)

4.1 Ingestion, for each file

1. **Read any format.** Detect format from magic bytes, not the extension. Detect encoding: BOM (UTF-16/ UTF-8), then UTF-8, then cp1252 with errors="replace". Unescape HTML entities (ORV-0001 and ORV-0022 contain "). Normalise CRLF.
2. **Parse the header** from the first text block (KEY: value lines up to ---). The in-file header wins. Cross-check against the manifest; log every mismatch.
3. **Split into paragraphs** on [P#]. Citations take the form DOC_ID#P#.
4. **Assign a health status:** ok, empty, truncated, encoding_damaged, duplicate_of:<id>, injection_flagged, header_mismatch.

    - Truncated = ends mid-sentence, or much shorter than its near-duplicate.
    - Injection = regex match (ASSISTANT:, SYSTEM:, "ignore your rules", "approve all", "skip cross") plus a cheap LLM classifier. Flagged paragraphs stay visible, are labelled "embedded instruction", and every agent sees document text wrapped in <document_data> tags.

5. **Derive sector** from the title or route line when it disagrees with the header (fixes ORV-0029 and ORV-0030).
6. **Parse trade data** into TradeRow {row_id, origin, dest, sector, month, value_usd_m, volume_t, unit, source_doc, validity}.
    - Prose notes: regex on "In (Month) (YYYY), the value was USD ([\d.]+)M and the volume was ([\d,]+) tonnes".
    - Fixed tables: header block, column line, numbered rows. Row ID is TABLE_ID:row.
    - Validate every row: missing value, non-numeric value, repeated month, cut-off table, value/volume ratio outside the ATL-0007 price band. **Flagged rows are shown but never used in calculations.**
7. **Build the correction graph.** Link each correction/retraction to the earlier claims it corrects (LLM plus date/ title matching): 0013 corrects 0043, which taints 0015. Also record superseding notices: a later Tier 1 notice overrides an earlier forecast.
8. **Live ingest.** POST /ingest (upload or folder watch) indexes incrementally and is idempotent by content hash. The final round depends on this.

4.2 Tools shared by the agents

- **search(query, filters, k)**: BM25 and dense scores fused with Reciprocal Rank Fusion. Filters: event_id, country, sector, source_type, max_tier, date_from (well above the two-filter minimum). **pub_date <= as_of is enforced inside the tool and cannot be overridden.** Results carry health and retracted_by. A case-wide query ledger rejects any repeat of the same normalised (query, filters) with DUPLICATE_QUERY.
- **read_doc(id, paras)**
- **get_trade_series(origin, dest, sector)**
- **compute_change(rows_a, rows_b, mode=value|volume, net_of=[...])**: a deterministic calculator, so no number in a claim is produced by the LLM itself.

5. The agents

	Heimdall (Gatekeeper)	Thor (Alarm)	Loki (Trickster)	Odin (Judge)
Tools	search, read_doc, get_trade_series, correction-graph lookup, write_case_file	search, read_doc, get_trade_series, compute_change, submit_claim, attack, defend, concede	same as Thor	read Case File, verify_citation, compute_change, rule, remand, write_briefing. **No search**, so he only judges.
Input	question	Case File	Case File	Case File + transcript
Output	Case File + verification verdict	claims with high bounds per country and sector	claims with low bounds	rulings, classification, ranges, remand or briefing

5.1 Heimdall

1. Parse the question: as-of date, countries [Midgard, Tolan], sectors [wheat, maize], event phrase, metric (import value change).
2. Resolve the event: search, cluster results by event_id, pick E5.
3. **Verification verdict:** REAL, NOT_REAL or UNCERTAIN, with reasons. REAL needs at least one Tier 1/2 confirmation that is not retracted (here ORV-0016 and ORV-0032). Also record fake sub-claims (0010, 0026) and retractions (0043, 0015). On NOT_REAL, stop and tell the user why. This is how fake-event packs get rejected.
4. Gather the evidence pack: documents and trade rows for every (country, sector) pair, plus a coverage map. In E5 the Tolan maize cell is empty, and that gap is written down here.
5. On remand, run **only new** queries targeting Odin's gap list. Memory lets it skip queries already run.

5.2 Thor and Loki (same structure, opposite mandates)

- **Round 1 (parallel, asyncio.gather):** each side runs at least 2 of its own searches and submits 3-6 claims.

- **Claim schema:** id, side, country, sector, statement; estimate {low, high, metric, baseline}; citations [{doc_id, para, pub_date, pub_time, quote}]; nature (fact or inference); reasoning. A claim without a citation is rejected by the schema. "Maybe it will be fine" is impossible because an estimate is mandatory.
- **Round 2:** each side files at least 1 attack, typed contradicts, retracted, low_tier, wrong_scope, stale or math_error, backed by its own evidence.
- **Round 3:** each side answers every attack it received with defend (must cite **new** evidence not used before) or concede.
- **Prompt rule for both:** "Conceding a claim the evidence defeats raises your credibility score with the judge." Odin tracks a per-side credibility score, so conceding is a real incentive.

5.3 Odin

1. **Deterministic checks first, then the LLM ruling, for every claim:** the citation exists in the manifest; pub_date <= as_of; the quote or number actually appears in the cited paragraph; the source is not retracted or tainted; source tier; numbers re-calculate correctly from the trade rows.
2. **Ruling per claim:** upheld, rejected, conceded or contested; nature sourced_fact or inference; a reason.
3. **Classification:**
    - **Established:** upheld, Tier 1/2, and survived an attack or was never credibly countered.
    - **Contested:** credible evidence on both sides, unresolved.
    - **Conceded:** withdrawn by its owner.
4. **Odin's range:** start from the upheld trade-row calculations, then narrow to where the surviving Thor and Loki bounds overlap. Stay inside the band the evidence supports and give a one-line reason for the width.
5. **Sufficiency rubric (a checklist, not a feeling):** every (country, sector) cell in the question has an Established claim with a trade row, or is explicitly "evidence missing"; both sides were challenged; no unresolved retraction; substitution (net effect) addressed. Any failure produces remand {missing: [...]}. **At least 1 remand; hard maximum 3 runs.**
6. **Briefing, written only from the rulings table** (he never sees raw argument text when writing):
    - Executive summary.
    - Established, Contested and Conceded findings.
    - Exposure table: Thor, Loki and Odin low/high per country and sector.
    - Next-quarter outlook with stated assumptions (e.g. port index stays near 30; rail capped per ORV-0050; no new agreement as of the as-of date).
    - Inline citations such as [ORV-0009#P5, 2027-01-19 09:20] and a source list.
    - "Unchallenged" warnings.
    - Data-quality appendix: damaged documents and rows, injection detected, clock exclusions.

5.4 How the second run brings new evidence honestly

Heimdall's first retrieval is focused and capped (top-k on the question's direct entities: Orvan to Midgard/Tolan). After run 1, Odin's gap list will usually read "net effect of substitute supply" and "Tolan maize". Heimdall's run-2 queries then surface ORV-0027, ORV-0044, ORV-0042 and ORV-0005, and confirm the Tolan maize gap from ATL-0010. That is genuinely new evidence, not staged.

6. Trial loop and hard limits

```
MAX_RUNS=3                          ROUNDS_PER_RUN=3 (claims, attacks, defences)
MAX_TOOL_CALLS_PER_AGENT_PER_RUN=12  MAX_TOKENS_PER_AGENT_PER_RUN=40k
MAX_CLAIMS_PER_SIDE=6               LLM_RETRIES=2 (exponential backoff)
CALL_TIMEOUT=60s                    CASE_WALLCLOCK=10min
DUPLICATE_QUERY=reject
```

- **If one side fails** (exception or limit): record side_failed and continue. The briefing must say "Thor's claims went unchallenged", and **an unchallenged claim can never become Established**; at best it is "uncontested".
- If a limit is hit while the case is still insufficient, the briefing is marked INCOMPLETE with reasons. Never claim a case is complete when it is not.

7. Private, lasting memory

- **Storage:** one SQLite file per agent (memory/<agent>.db), persisting across cases. No agent can read another's store; only the Case File is shared.
- **Heimdall** remembers queries already run per case (blocks repeats) and observed source reliability (e.g. "Raven Wire issued a correction").
- **Thor and Loki** remember their conceded claims and why, which attacks worked, and which sources were knocked down (Loki learns "Vantro statements get contradicted").
- **Odin** remembers precedents, e.g. "uncorroborated Tier 1 political quote vs trade data: rejected".
- **Every read is logged** as memory_read {key, value, influenced: "skipped re-asserting claim T-4"}. That line is the proof for acceptance criterion 9. Example to demo: in run 2 Thor does *not* re-submit the Ledger 90% claim because his memory records he conceded it.

8. Observability and reproducibility

- **Event log** cases/<case_id>/log.jsonl, one event per line: query, retrieval, claim, attack, defence, concession, memory_read, memory_write, ruling, remand, doc_health, row_flagged, injection_detected, clock_excluded, limit_hit, error.
- **Per run:** run_<n>/case_file.json, rulings.json, transcript.json.
- **Per case:** briefing.md, briefing.json, and meta.json (as-of date, manifest hash, index hash, model IDs, config, git SHA).
- **replay command** rebuilds the briefing from cached LLM responses.
- **Submission:** commit one full E5 case with 2 runs.

9. Dashboard

1. **Question box** with an as-of date picker.
2. **Live feed:** documents replayed in pub_date/pub_time order up to the as-of date, with badges for tier, retracted, damaged and injection.
3. **Trial view:** three columns (Thor, Odin's rulings, Loki), grouped by run and round. Attack arrows link claims; conceded claims are struck through.
4. **Exposure table:** country and sector rows with Thor, Loki and Odin ranges as bars on one axis. "Evidence missing" cells (Tolan maize) clearly marked.
5. **Click-through:** every citation opens the document viewer at the highlighted paragraph; trade citations open the series table at the highlighted row.
6. **Briefing tab** and **Trace tab** (filterable JSONL log, with a run 1 vs run 2 diff showing the new evidence).
7. **Ingest button** for the final round.

10. Build order

Phase	Deliverable	Done when
1. Data layer	Ingest (all formats, health checks), header and trade parsers, validation, hybrid index, clock filter, query ledger	Unit tests cover every trap in 2.1: ORV-0023 empty, ORV-0007 encoding, 0029/0030 sector fix, 0014/0036/0041 excluded, ORV-0020 flagged, duplicates collapsed
2. Case File + Heimdall	Pydantic schemas, verification verdict, correction graph	E5 is REAL with 0043 and 0015 retracted and 0010/0026 rejected; a fake-event test question returns NOT_REAL with reasons
3. Thor and Loki	Tool-using agents, claim schema, parallel round 1, attack and defend/concede rounds	Each runs its own searches; at least 1 concession on E5 (0004, 0033 or 0045)
4. Odin + loop	Deterministic citation checks, rulings, classification, range synthesis, sufficiency rubric, remand, limits	A 2-run E5 case completes and run-2 evidence differs from run 1

5. Memory + logs	Per-agent stores, logged memory influence, replay	Acceptance criterion 9 visible in the log
6. Dashboard	The 7 panels in section 9, live via SSE	Click-through works for documents and trade rows
7. Final-round hardening	Holdout drill: remove all E5 docs from the index, start the server, ingest E5 live, ask the question. Fuzz damaged files (cut-off table, repeated row, "12,O00" bad number)	Runs end to end with no code edits and no crashes
8. Submission	README, .env.example, architecture diagram, saved 2-run logs	Everything in PS section 3.4 is present

If time runs short, cut in this order: embedding search (keep BM25 + filters, and add a cheap TF-IDF cosine so RRF is still hybrid), live-feed animation polish, the credibility score.
Never cut: deterministic citation and number checks, the clock filter, damaged-row flagging. These are scored directly.

11. Acceptance criteria map

#	Criterion (PS 3.3)	Where it is met
1	Ingest PDF/TXT/DOCX incl. live docs; Case File; trade tables; event check	4.1, 5.1, POST /ingest
2	Hybrid search with at least 2 metadata filters	4.2 search()
3	Thor and Loki submit evidence-backed claims from their own searches	5.2 Round 1
4	At least one cross-examination	5.2 Rounds 2-3
5	Odin rules on every claim; at least one rejected or conceded	5.3; traps 0004 / 0033 / 0045
6	Briefing: Established/Contested/Conceded, ranges, timestamped citations, outlook	5.3 step 6
7	Dashboard: question box, feed, trial view, exposure table, click-through	Section 9
8	At least one remand and a second run with new evidence	5.3 step 5, 5.4
9	Private lasting memory changes an agent's action (logged)	Section 7
10	Failure case handled; hard limits on rounds, runs, retries	4.1 health status, section 6

12. Evaluation harness

eval/score.py runs over a saved case and checks:

- Share of citations that resolve to the manifest.
- Clock violations (must be 0).
- Retracted or tainted documents used as Established (must be 0).
- Injection obeyed (must be 0).
- Repeated queries (must be 0).
- At least 1 concession and at least 1 remand.
- Whether the section 2.2 reference numbers fall inside each side's range, and the width of Odin's range.

Build 2-3 synthetic mini-packs (a fake event; a fixed-layout trade table with damaged rows) to test generalisation before the final round.

13. Repository fixes to make now

- **git core.autocrlf is true, and 58 of 76 files fail hashes.sha256.** Git has most likely rewritten line endings, which changes file bytes. Add a .gitattributes with "dev_pack_E5/** -text" and re-checkout, or verify against the original download, so the integrity check does not flag healthy files.

- **The README forbids passing runemark_map.png to agents.** Ingest should read only what manifest.json lists.